

Assignment – 7

Question – 1

Code:

```
1/*
2Take a number from user in the main function. A function called
3factorial () that is invoked in main () function passing the number
4by value. The factorial () function must return the factorial value
5of the number passed. The factorial value returned from main ()
6function must be printed in main () function.
7*/
8
9#include <stdio.h>
10int factorial(int);
11int main() {
12    int num;
13    printf("Enter a number: ");
14    scanf("%d", &num);
15    printf("The factorial of %d is %d.\n", num, factorial(num));
16}
17
18int factorial(int n) {
19    int res = 1;
20    for(int i=n; i>0; i--) res*=i;
21    return res;
22}
```

Output:

```
students@cse1ab1admin-HP-Elite-SFF-600-G9-Desktop-PC: ~/Desktop/24CSE1027/assignment_7$ ./q1
Enter a number: 7
The factorial of 7 is 5040.
```

Question – 2

Code:

```
1/*
2Let there be 4 functions written corresponding to the arithmetic
3operations: addNum (), subNum(), mulNum() and divNum(). The
4main () function reads two numbers from user and the choice of
5the arithmetic operation that user desires to apply on two
6numbers. Accordingly the corresponding function must be
7invoked and result must be returned to main () function where it
8is displayed.[Here assume that situation of divide by zero case is
9verified before calling divNum() function]
10*/
11
12#include <stdio.h>
13float addNum(float x, float y) { return (x+y); }
14float subNum(float x, float y) { return (x-y); }
15float mulNum(float x, float y) { return (x*y); }
16float divNum(float x, float y) { return (x/y); }
17int main() {
18    float a, b;
19    int op, repeat=1;
20    while(repeat == 1) {
21        printf("\nEnter two numbers: ");
22        scanf("%f %f", &a, &b);
23        printf("[1] Addition\n[2] Subtraction\n[3] Multiplication\n[4] Division\nEnter the arithmetic operation: ");
24        scanf("%d", &op);
25        switch(op) {
26            case 1:
27                printf("The addition of %.2f and %.2f is: %.2f.\n", a, b, addNum(a, b)); break;
28            case 2:
29                printf("The subtraction of %.2f and %.2f is: %.2f.\n", a, b, subNum(a, b)); break;
30            case 3:
31                printf("The multiplication of %.2f and %.2f is: %.2f.\n", a, b, mulNum(a, b)); break;
32            case 4:
33                if(b == 0) printf("Division by zero is not allowed.\n");
34                else printf("The division of %.2f and %.2f is: %.2f.\n", a, b, divNum(a, b));
35                break;
36            default:
37                printf("Invalid operation.\n"); break;
38        }
39    }
40    printf("\nRepeat the program?\n [0] No [1] Yes: ");
41    scanf("%d", &repeat);
42}
43
```

Output:

```
students@cse1abiadn1n-HP-Elite-SFF-600-G9-Desktop-PC: ~/Desktop/24CSE1027/assignment_7$ ./q2
Enter two numbers: 4 8
[1] Addition
[2] Subtraction
[3] Multiplication
[4] Division
Enter the arithmetic operation: 4
The division of 4.00 and 8.00 is: 0.50.

Repeat the program?
[0] No [1] Yes: 1

Enter two numbers: 9 5
[1] Addition
[2] Subtraction
[3] Multiplication
[4] Division
Enter the arithmetic operation: 1
The addition of 9.00 and 5.00 is: 14.00.

Repeat the program?
[0] No [1] Yes: 0
```

Question – 3

Code:

```
1/*
2To use the built in mathematical functions sin (), cos(), tan(),
3pow(), sqrt() from math.h. Write a menu driven program that calls
4any of these functions based on user choice
5*/
6
7#include "stdio.h"
8#include "math.h"
9
10int main() {
11    int op, a, b, repeat=1;
12    while(repeat == 1) {
13        printf("\n[1] Calculate sine\n[2] Calculate cosine\n[3] Calculate tangent\n[4] Calculate power\n[5] Calculate square root\nEnter the operation to perform: ");
14        scanf("%d", &op);
15        if(op == 4) {
16            printf("Enter the base and the exponent respectively: ");
17            scanf("%d %d", &a, &b);
18            printf("The value of %d*%d is %d.\n", a, b, (int)pow(a, b));
19        } else {
20            printf("Enter a number: ");
21            scanf("%d", &a);
22            switch(op) {
23                case 1:
24                    printf("The value of sin(%d) is %.5f.\n", a, sin(a)); break;
25                case 2:
26                    printf("The value of cos(%d) is %.5f.\n", a, cos(a)); break;
27                case 3:
28                    printf("The value of tan(%d) is %.5f.\n", a, tan(a)); break;
29                case 5:
30                    printf("The value of sqrt(%d) is %.5f.\n", a, sqrt(a)); break;
31                default:
32                    printf("Invalid operation\n"); break;
33            }
34        }
35        printf("\nRepeat the program?\n [0] No [1] Yes: ");
36        scanf("%d", &repeat);
37    }
38 }
39 ]
```

Output:

```
students@cse1abiadn1n-HP-Elite-SFF-600-G9-Desktop-PC: ~/Desktop/24CSE1027/assignment_7$ ./q3
[1] Calculate sine
[2] Calculate cosine
[3] Calculate tangent
[4] Calculate power
[5] Calculate square root
Enter the operation to perform: 3
Enter a number: 90
The value of tan(90) is -1.99520.

Repeat the program?
[0] No [1] Yes: 1

[1] Calculate sine
[2] Calculate cosine
[3] Calculate tangent
[4] Calculate power
[5] Calculate square root
Enter the operation to perform: 5
Enter a number: 9
The value of sqrt(9) is 3.00000.

Repeat the program?
[0] No [1] Yes: 0
```

Question – 4 (by Reference)

Code:

```
1/*
2To swap the values of two numbers in a function called
3swapNumbers(). The main must call this function. The values of
4the numbers must be printed in main () before and after calling
5swapNumbers() function. The numbers must be printed in the
6swapNumbers() function before returning from function.
7*/
8
9// Call by Reference
10#include <stdio.h>
11
12void swapNumbers(int *, int *);
13int main() {
14    int a, b;
15    printf("Enter the values of a and b: ");
16    scanf("%d %d", &a, &b);
17    printf("Before calling swapNumbers(): a = %d, b = %d.\n", a, b);
18    swapNumbers(&a, &b);
19    printf("After calling swapNumbers(): a = %d, b = %d.\n", a, b);
20}
21
22void swapNumbers(int *x, int *y) {
23    int t = *x;
24    *x = *y;
25    *y = t;
26    printf("During calling swapNumbers(): a = %d, b = %d.\n", *x, *y);
27}
```

Output:

```
students@cseLabIadmn-HP-Elite-SFF-600-Q9-Desktop-PC:~/Desktop/24CSE1027/assignment_7$ ./q4_r
Enter the values of a and b: 4 6
Before calling swapNumbers(): a = 4, b = 6.
During calling swapNumbers(): a = 6, b = 4.
After calling swapNumbers(): a = 6, b = 4.
```

Question – 4 (by Value)

Code:

```
1/*
2To swap the values of two numbers in a function called
3swapNumbers(). The main must call this function. The values of
4the numbers must be printed in main () before and after calling
5swapNumbers() function. The numbers must be printed in the
6swapNumbers() function before returning from function.
7*/
8
9// Call by Value
10#include <stdio.h>
11
12void swapNumbers(int, int);
13int main() {
14    int a, b;
15    printf("Enter the values of a and b: ");
16    scanf("%d %d", &a, &b);
17    printf("Before calling swapNumbers(): a = %d, b = %d.\n", a, b);
18    swapNumbers(a, b);
19    printf("After calling swapNumbers(): a = %d, b = %d.\n", a, b);
20}
21
22void swapNumbers(int x, int y) {
23    int t = x;
24    x = y;
25    y = t;
26    printf("During calling swapNumbers(): a = %d, b = %d.\n", x, y);
27}
```

Output:

```
students@cseLabIadmn-HP-Elite-SFF-600-Q9-Desktop-PC:~/Desktop/24CSE1027/assignment_7$ ./q4_v
Enter the values of a and b: 4 6
Before calling swapNumbers(): a = 4, b = 6.
During calling swapNumbers(): a = 6, b = 4.
After calling swapNumbers(): a = 4, b = 6.
```

Question – 5

Code:

```
1/*
2Write a function called isPrime() that returns 1 if the number
3passed from main() function is a prime number.
4*/
5
6#include <stdio.h>
7
8int isPrime(int);
9int main() {
10    int num;
11    printf("Enter a number: ");
12    scanf("%d", &num);
13    printf("Value returned by the isPrime() function is: %d.\n", isPrime(num));
14}
15
16int isPrime(int n) {
17    int is = 1;
18    if(n < 2) { printf("Invalid number provided.\n"); is = 0; }
19    for(int i=2; i<n; i++) {
20        if(n%i==0) is=0;
21    }
22    return is;
23}
```

Output:

```
students@cseabladn-in-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_7$ ./q5
Enter a number: 6
Value returned by the isPrime() function is: 0.
students@cseabladn-in-HP-Elite-SFF-600-G9-Desktop-PC:~/Desktop/24CSE1027/assignment_7$ ./q5
Enter a number: 7
Value returned by the isPrime() function is: 1.
```